Enrolment Number

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Winter, 2025

B.Tech 4th Semester Examination

Civil Engineering-Societal and Global Impact

Course Code: MNCAC2

Full Marks – 50 **Time – 2 hours**

The figure in the margin indicates full marks for the questions.

SECTION 1

Answer all the questions **Marks: 10 x 1 = 10**

1(a) Which of the following best describes the primary aim of civil engineering in society?

i) Maximizing profit
ii) Improving society and well-being
iii) Abiding by government policies
iv) Promoting private sector growth

1(b) Which of the following best quantifies the carbon footprint of a civil engineering project?

i) Total cost of construction
ii) Sum of direct and indirect greenhouse gas emissions over the project life-cycle
iii) Area of land used
iv) Number of workers employed

1(c) The resilience of a transportation system is best measured by its ability to:

i) Maintain functionality during and after disruptive events
ii) Minimize construction costs
iii) Maximize vehicle throughput
iv) Reduce number of signals

1(d) The main advantage of BIM (Building Information Modeling) in infrastructure is:

i) Better 3D visualization
ii) Integrated project delivery and lifecycle management
iii) Faster concrete curing
iv) Lower steel consumption

1(e) In a smart city, which technology is most used for real-time traffic management?

i) GIS-based mapping
ii) IoT sensors and adaptive signal control
iii) Traffic policing
iv) CCTV only

1(f) Which is the most effective strategy for flexible space management in office buildings?

i) Fixed cubicle partitions
ii) Modular furniture and movable walls
iii) Single open hall
iv) None of the above

1(g) Which renewable energy source is most suitable for decentralized rural electrification in India?

i) Large hydroelectric power plants
ii) Solar photovoltaic
iii) Nuclear plants
iv) Diesel generators

1(h) Which of the following best demonstrates the concept of circular economy in construction?

i) Landfilling demolition waste
ii) Reusing and recycling construction materials
iii) Importing new cost-effective materials
iv) Burning waste on site

1(i) The best approach for retrofitting old buildings for energy efficiency is:

i) Adding insulation and upgrading windows
ii) Repainting walls
iii) Increasing floor area
iv) Planting vegetationinside homes

1(j) Which of the following is a unique feature of the GRIHA rating system compared to LEED?

i) Focus on international building codes
ii) Emphasis on Indian climatic conditions and local construction practices
iii) Mandatory use of imported materials
iv) Exclusive assessment of energy efficiency

SECTION 2

Answer any one question from this section (300 words) **Marks: 10**

2(a) Explain the concept of life cycle assessment (LCA) and illustrate its importance in sustainable civil engineering. (5+5)

2(b) Explain the concept of facility management and demonstrate its role in addressing climate change. (5+5)

SECTION 3

Answer any one question from this section (300 words) **Marks: 10**

3(a)Identify the basic limitations of large dams. Utilize your knowledge of civil engineering techniques to suggest practical solutions for overcoming or minimizing those limitations.

3(b) Apply your knowledge of civil engineering techniques to model the construction of any famous building, monument or infrastructure of your choice.

SECTION 4

Answer any one question from this section (300 words) **Marks: 10**

4(a) Analyze the concept of infrastructure in the context of civil engineering. Distinguish between the various types of critical infrastructure – transportation, energy, water resources management and telecommunication. (5+5)

4(b) List the various components of a building control system with special emphasis on security systems and active/passive design strategies. (4+3+3)

SECTION 5

Answer any one question from this section (300 words) **Marks: 10**

5(a) Critically assess the concept of sustainability and justify how civil engineering projects can contribute to sustainable development. (6+4)

5(b) Critically evaluate the role of civil engineers in promoting resilience and adaptability in the face of globa environmental challenges.